

Reminders

Upcoming due dates

Wed Mar 04 EDA checkpoint

Fri Mar 06 D8

Natural Language Processing (NLP)

Data Science in Practice

Jason G. Fleischer, PhD

Dept. of Cognitive Science

UC San Diego

Examples of questions that require text analysis

1. Did J.K. Rowling write The Cuckoo's Calling under the pen name Robert Galbraith?
2. What themes are common in 19th century literature?
3. Can we tell the difference between tweets that come from Trump himself or a staffer?
4. Is Hillary the most poisoned name in US History?
5. Can we visualize the narrative structure of The Hobbit?
6. Who has the biggest vocabulary in hip hop?
7. Is there a gender imbalance in who gets to say lines in the movies?

A whole bunch of text processing project ideas

- Spam detection (in text messages, on social media)
- Hate speech detection
- Predictive text (like gmail)
- Summarize articles (TL;DR)
- Summarize trends on social media... or in cutting edge NLP research?
- Plagiarism detection... in code projects?
- Document similarity... prevent the posting of duplicate questions on Campuswire?

Today's example question: How has pop music changed in the last five years?

Goal: Understand the basics of sentiment analysis and TF-IDF

What data would we need to answer this question?

How has pop music changed in the last five years?

Data: Lyrics to the most popular songs
from each year

The data : Top songs from Feb music charts 2017-2021

2017: 152 songs

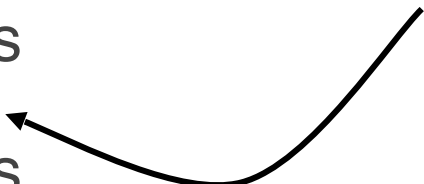
2018: 139 songs

2019: 127 songs

2020: 137 songs

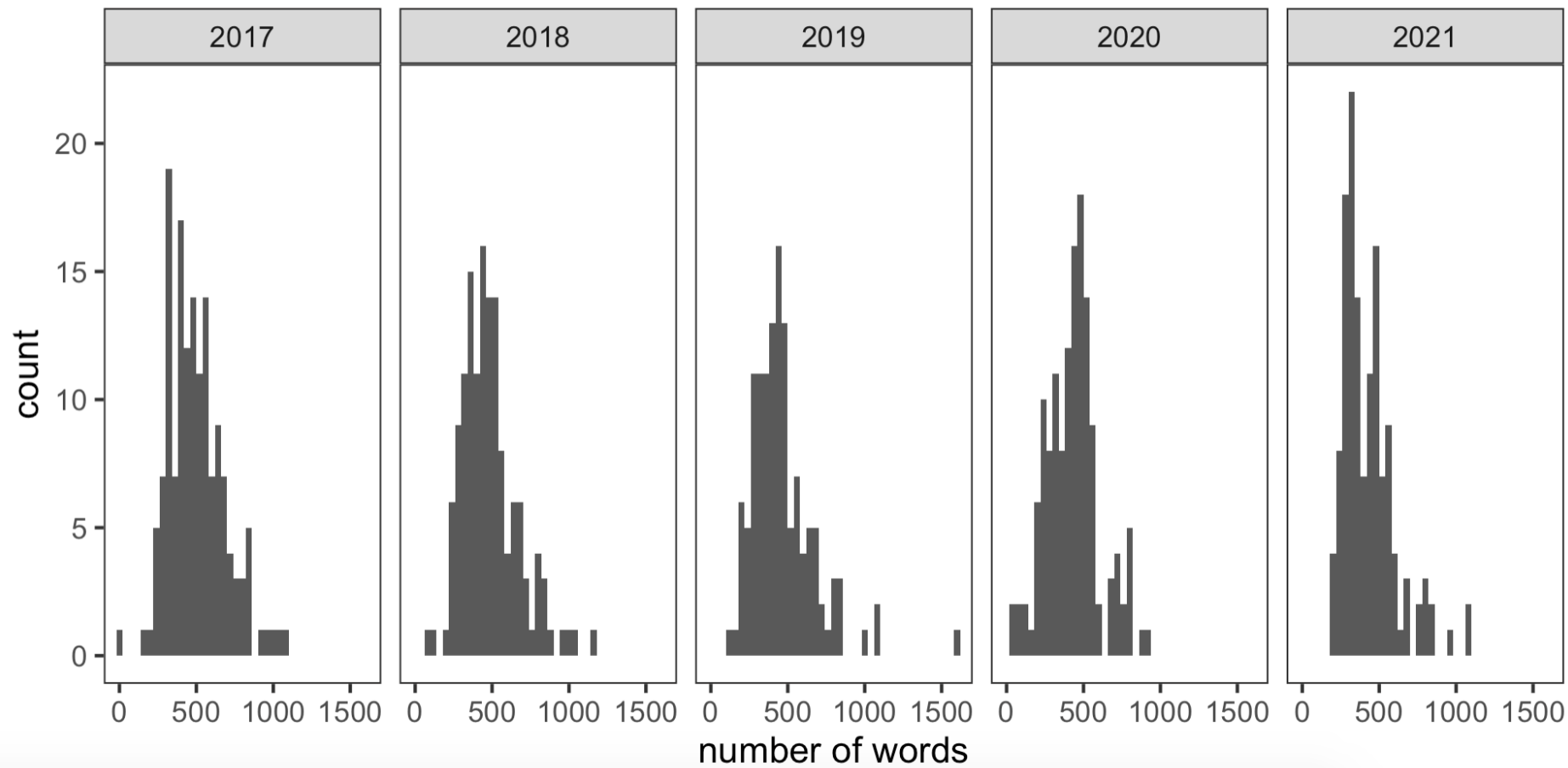
2021: 134 songs

Song data from **Spotify**.
Lyrics from **genius.com**

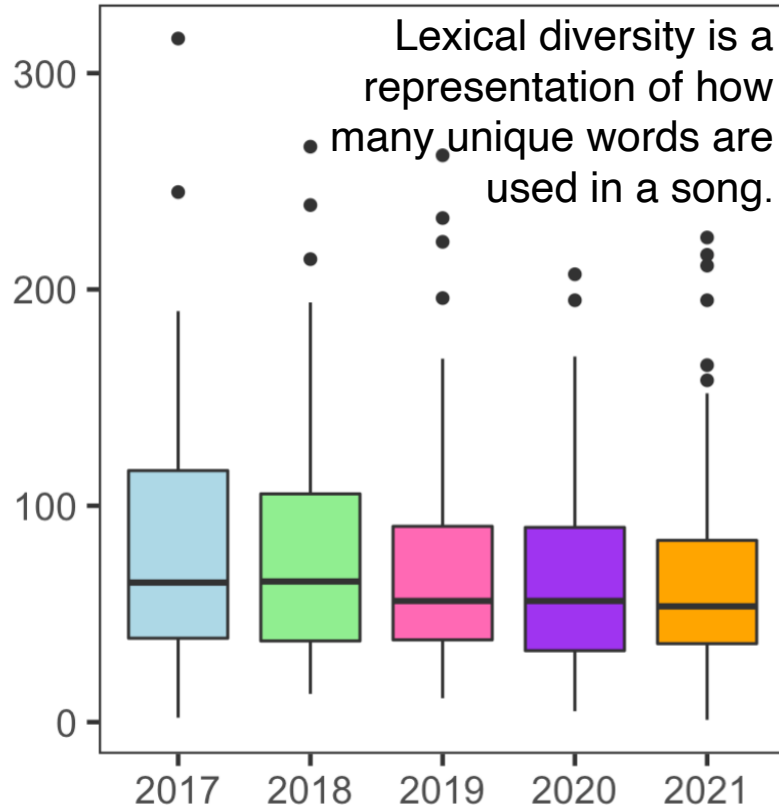
A curved arrow originates from the text 'Song data from Spotify. Lyrics from genius.com' and points towards the year '2021' in the list of songs.

Questions we can ask...

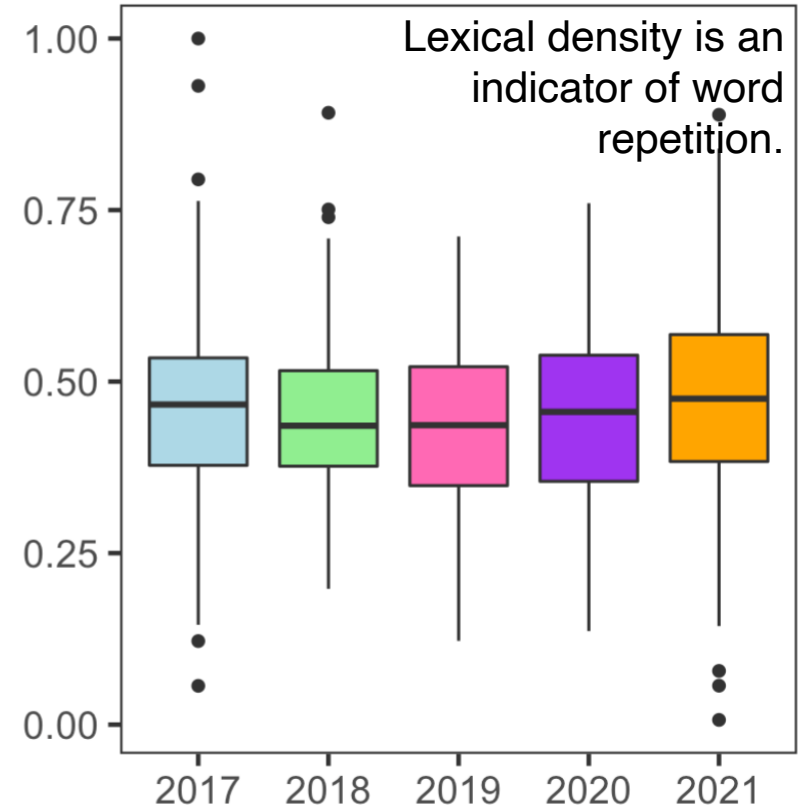
1. Does the total number of words change over time?
2. Does uniqueness change over time?
3. Does the diversity or density change?
4. What words are most common?
5. What words are most unique to each year?
6. What sentiment do songs convey most frequently?
7. Has sentiment changed over time?
8. What are the sentiment of the #1 songs?
9. What words contribute to the sentiment of these #1 songs?
- 10....what about bigrams? N-grams?



Lexical Diversity



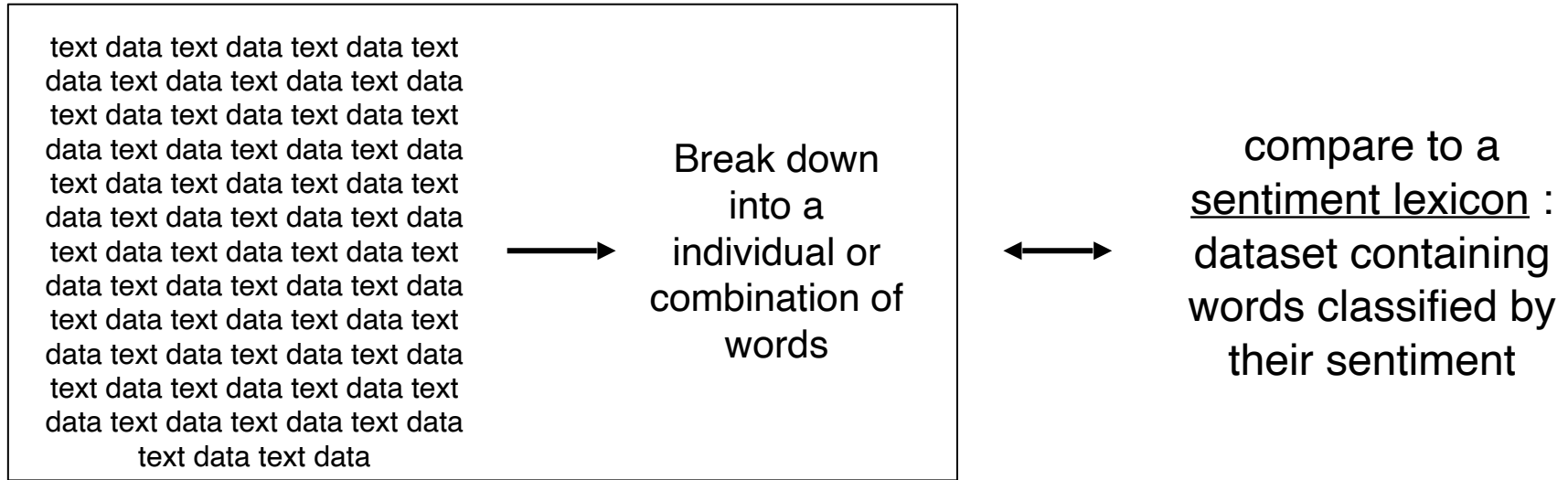
Lexical Density



Sentiment Analysis

Sentiment Analysis

Programmatically infer emotional content of text



Part of the
“NRC”
sentiment
lexicon:

word	sentiment	lexicon
<chr>	<chr>	<chr>
abacus	trust	nrc
abandon	fear	nrc
abandon	negative	nrc
abandon	sadness	nrc
abandoned	anger	nrc
abandoned	fear	nrc
abandoned	negative	nrc
abandoned	sadness	nrc
abandonment	anger	nrc
abandonment	fear	nrc
... with 27,304 more rows		

When doing sentiment analysis...

token - a meaningful unit of text

Check out <https://neptune.ai/blog/tokenization-in-nlp>

- what you use for analysis
- *tokenization* takes corpus of text and splits it into tokens (words, bigrams, etc.)

stop words - words not helpful for analysis

- extremely common words such as “a”, “the”, “of”, “to”
- are typically removed from analysis

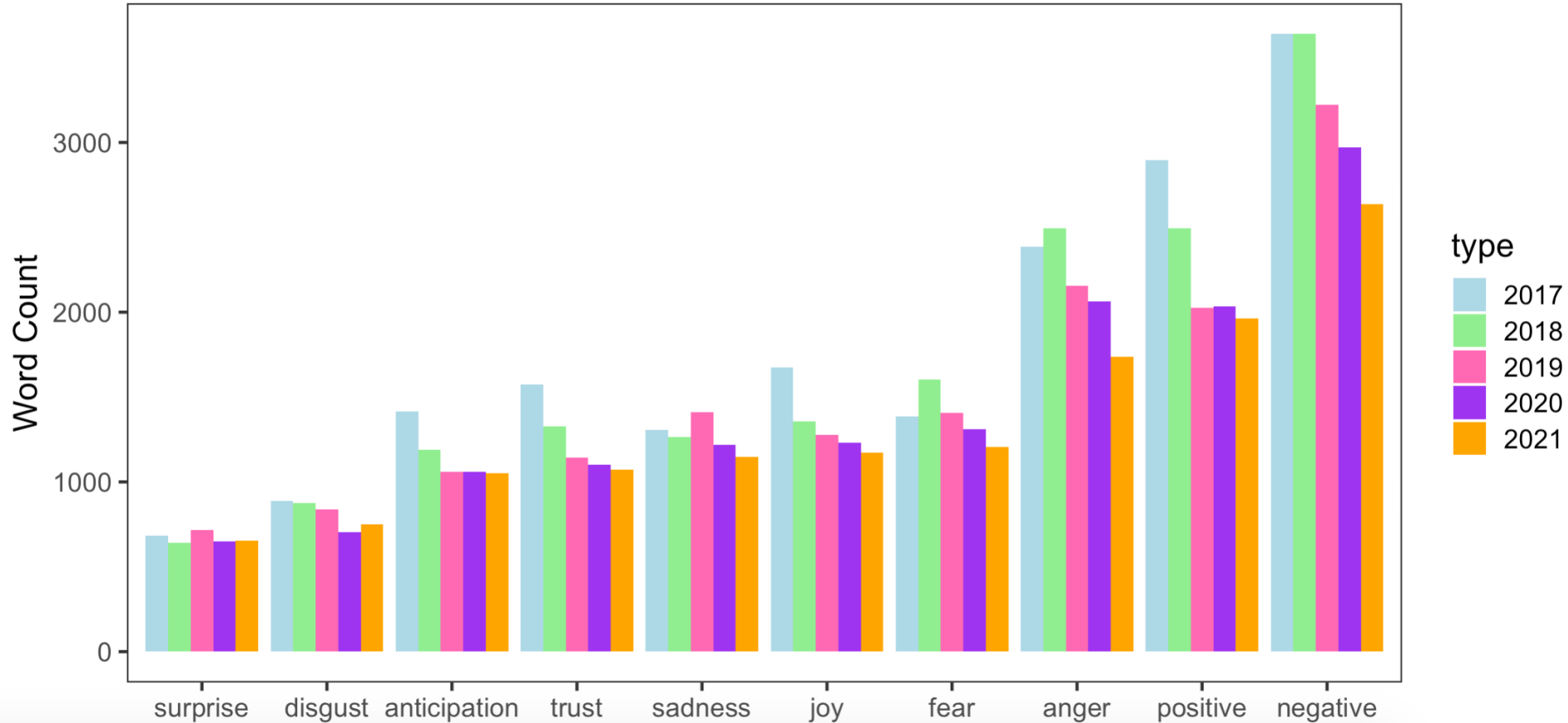
When doing sentiment analysis...

stemming or lemmatization

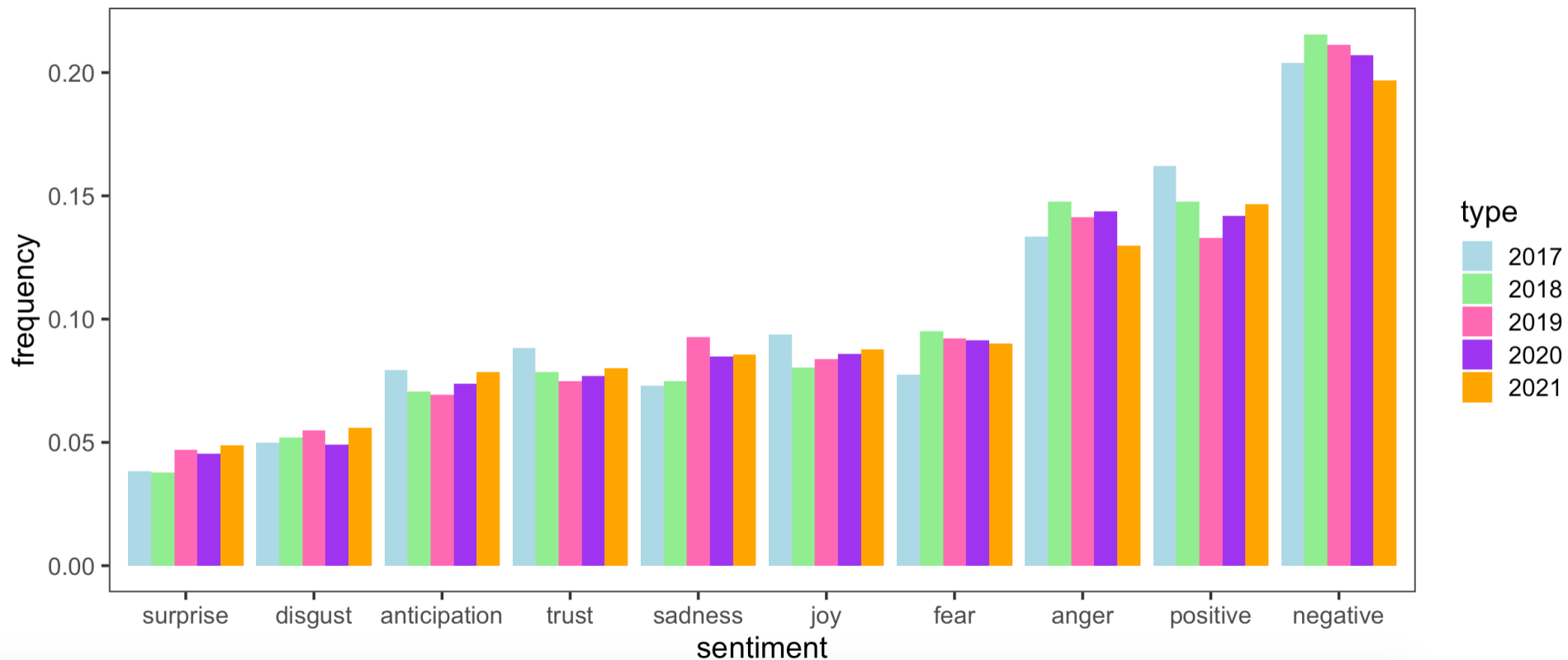
- Identifying the root for each token
- Jumping, jumped, jumps, jump all have the same root 'jump'
- Where things get tricky: jumper???
- Stemmers and lemmatizers do the same thing, stemmers are word focussed/cruider and lemmatizers try to take into account the context to correctly stem jumper differently depending on if its "I love my new jumper" (In the US jumper=a kind of dress, in the UK jumper=sweater) or its "Shane is a long jumper"

Check out <https://www.datacamp.com/tutorial/stemming-lemmatization-python>

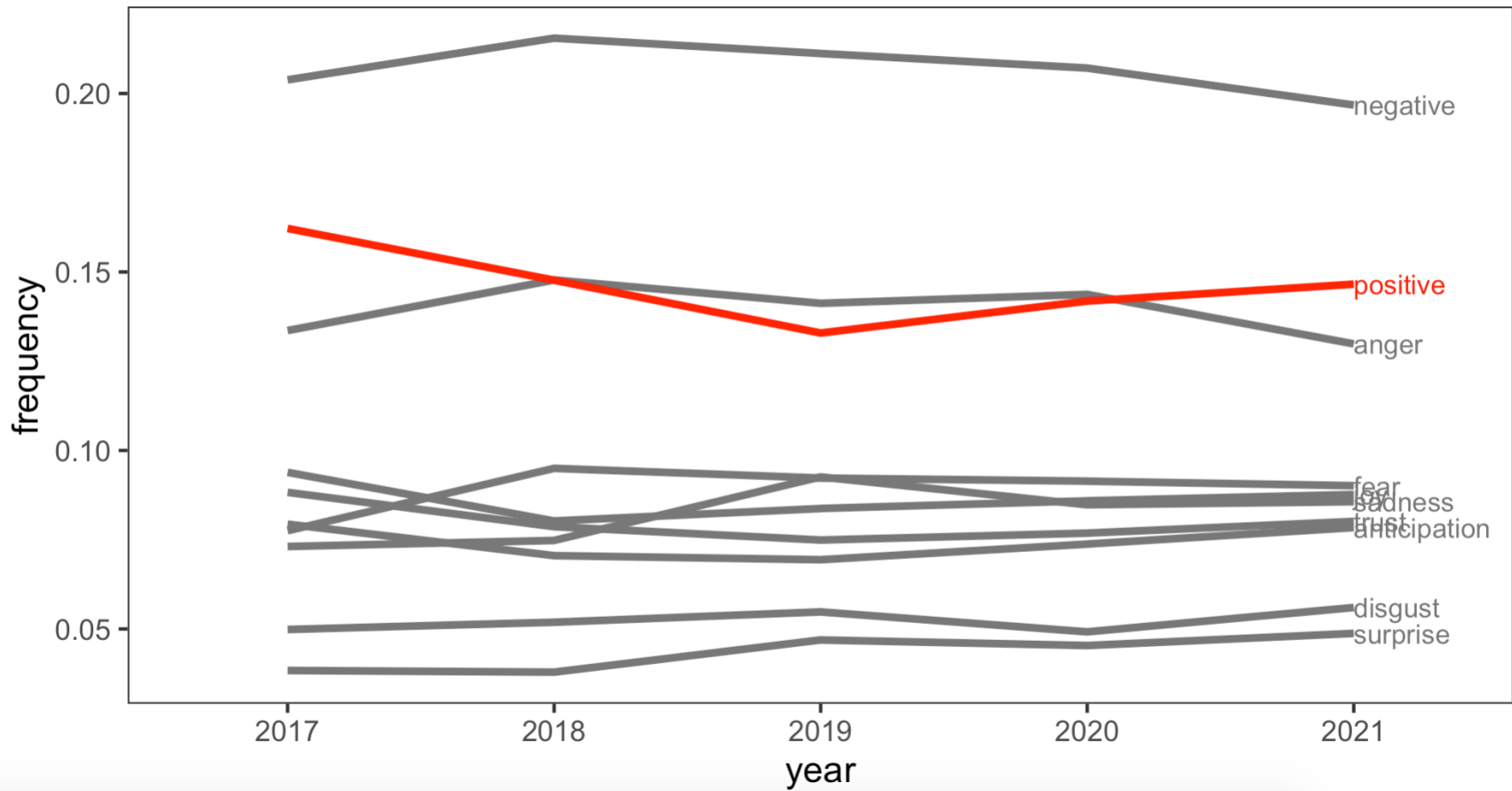
Top Songs Sentiment



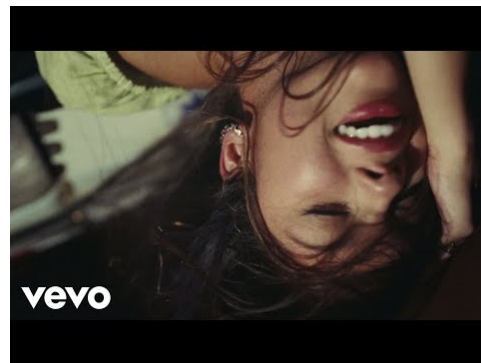
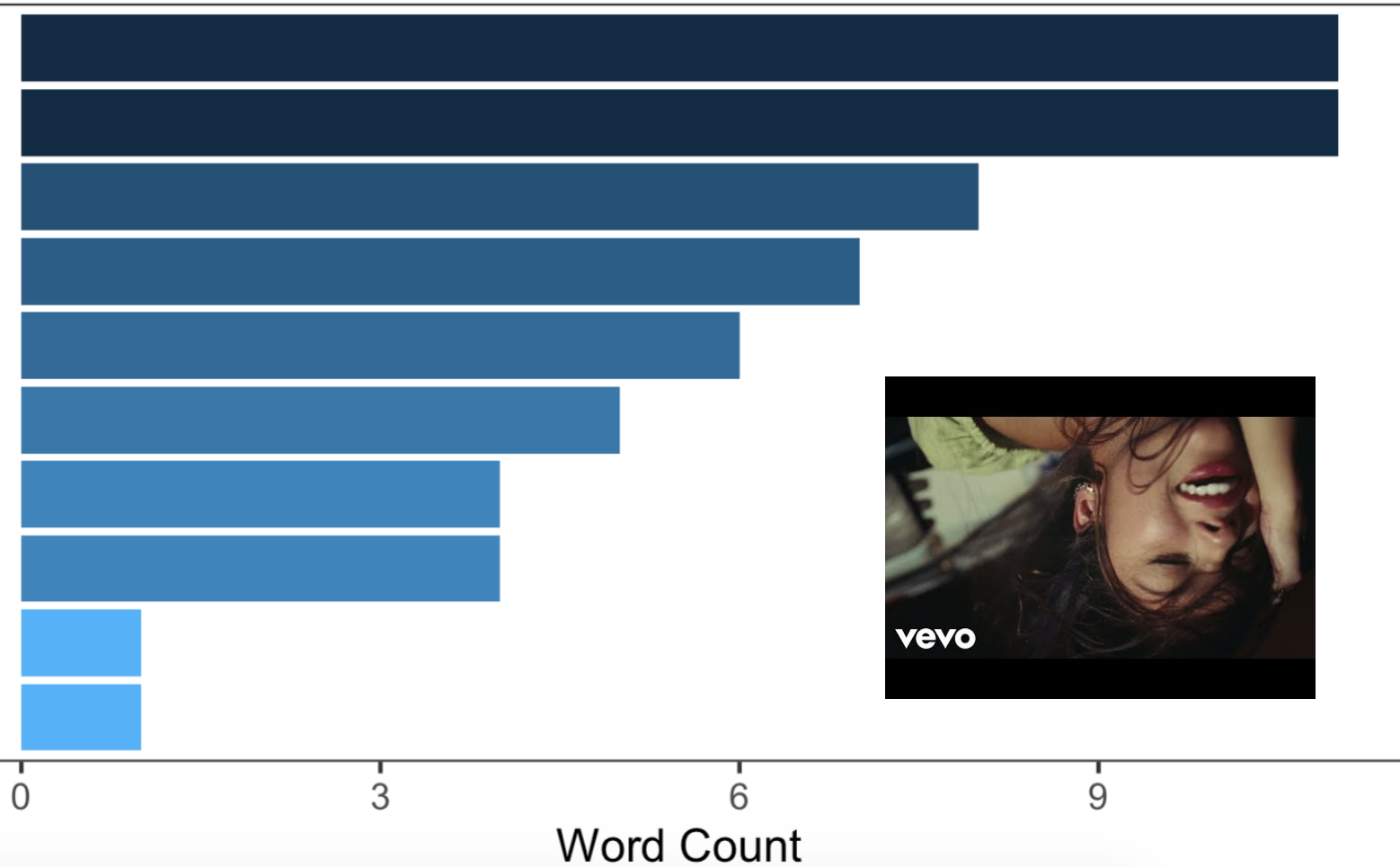
Sentiment by Year



Change in Sentiment over Time



Sentiment: Driver's License



I got my driver's license last week
Just like we always talked about
'Cause you were so excited for me
To finally drive up to your house
But today I drove through the suburbs
Crying 'cause you weren't around
And you're probably with that blonde girl
Who always made me doubt
She's so much older than me
She's everything I'm insecure about
Yeah, today I drove through the suburbs
'Cause how could I ever love someone else?
And I know we weren't perfect but I've never felt this way for no one
And I just can't imagine how you could be so okay now that I'm gone
Guess you didn't mean what you wrote in that song about me
'Cause you said forever, now I drive alone past your street
And all my friends are tired
Of hearing how much I miss you, but
I kinda feel sorry for them
'Cause they'll never know you the way that I do, yeah
Today I drove through the suburbs
And pictured I was driving home to you



Sentiment Limitations

How would you classify the sentiment of the following sentence?

“The idea behind the movie was great, but it could have been better”

A positive B negative C neutral D other

“Dude!”

“So bad!”

The limitations of sentiment analysis:

Context, homonyms, pragmatics, parts of speech

Mid, Shade

The limitations of sentiment analysis:

Lexicon not up to the task, lexicon specific results

TF-IDF:

Term frequency - Inverse document frequency

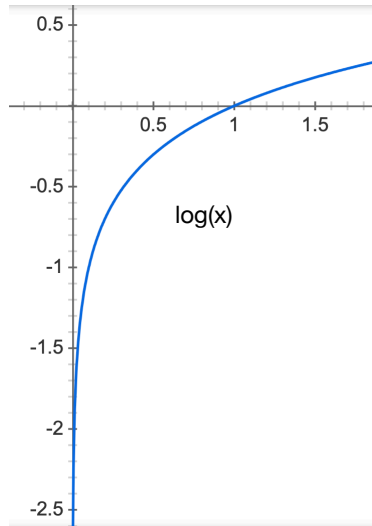
Given a corpus of documents, figure out which words are uniquely important to each document

TF-IDF:

Term Frequency - Inverse Document Frequency

Term Frequency (TF) : how frequently a word occurs in a document

Inverse document frequency (IDF) : how important a word is to a document



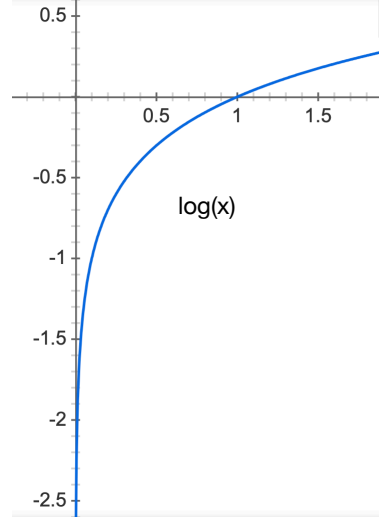
$$idf(\text{term}) = \ln \left(\frac{n_{\text{documents}}}{n_{\text{documents containing term}}} \right)$$

decreases the weight for commonly used words and
increases the weight for words that are not used very much
in a collection of documents

TF-IDF:

Term Frequency - Inverse Document Frequency
the frequency of a term adjusted for how rarely it is used

$$w_{x,y} = tf_{x,y} \times \log \left(\frac{N}{df_x} \right)$$



TF-IDF

Term x within document y

$tf_{x,y}$ = frequency of x in y

df_x = number of documents containing x

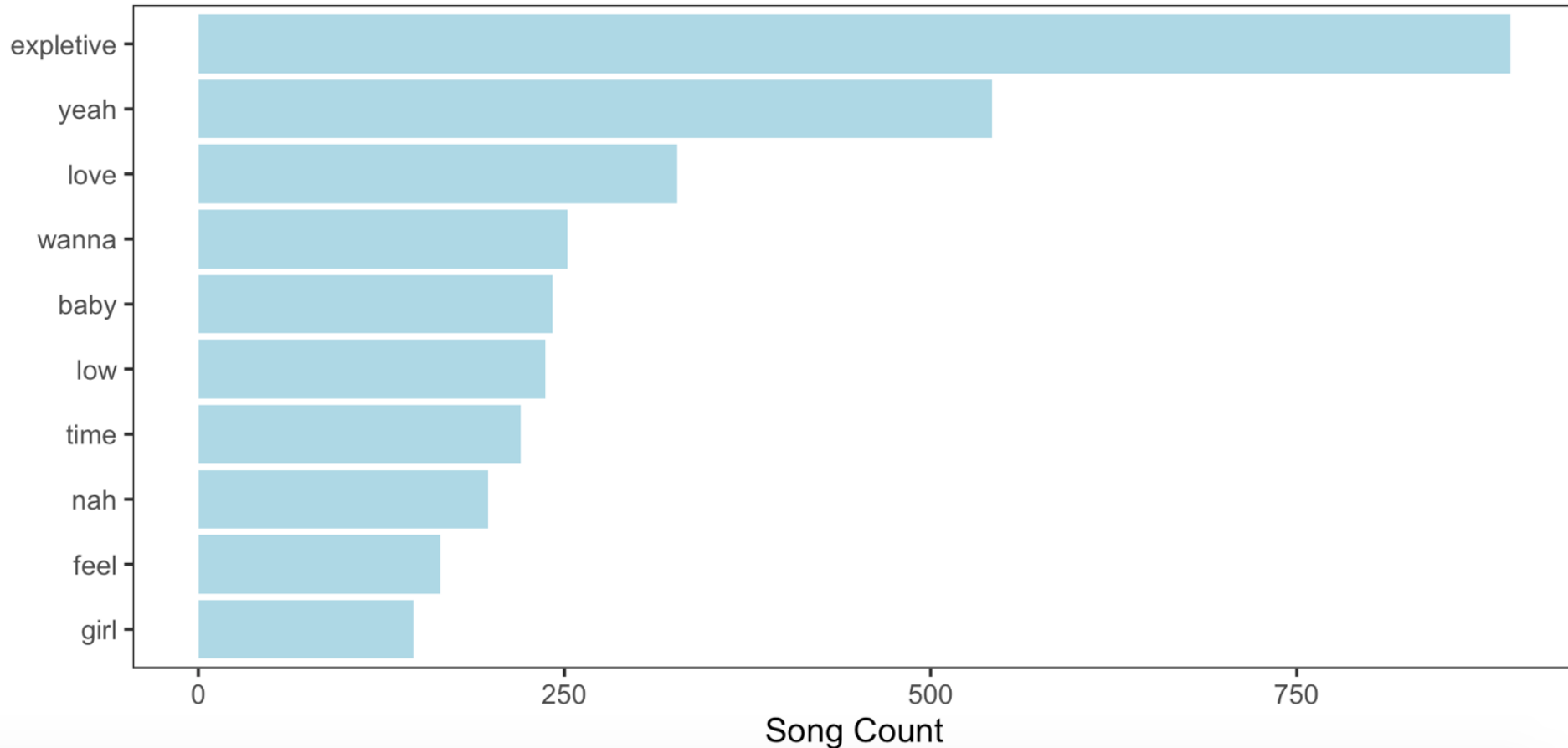
N = total number of documents

What words are the most unique to the lyrics of each year's top hits?

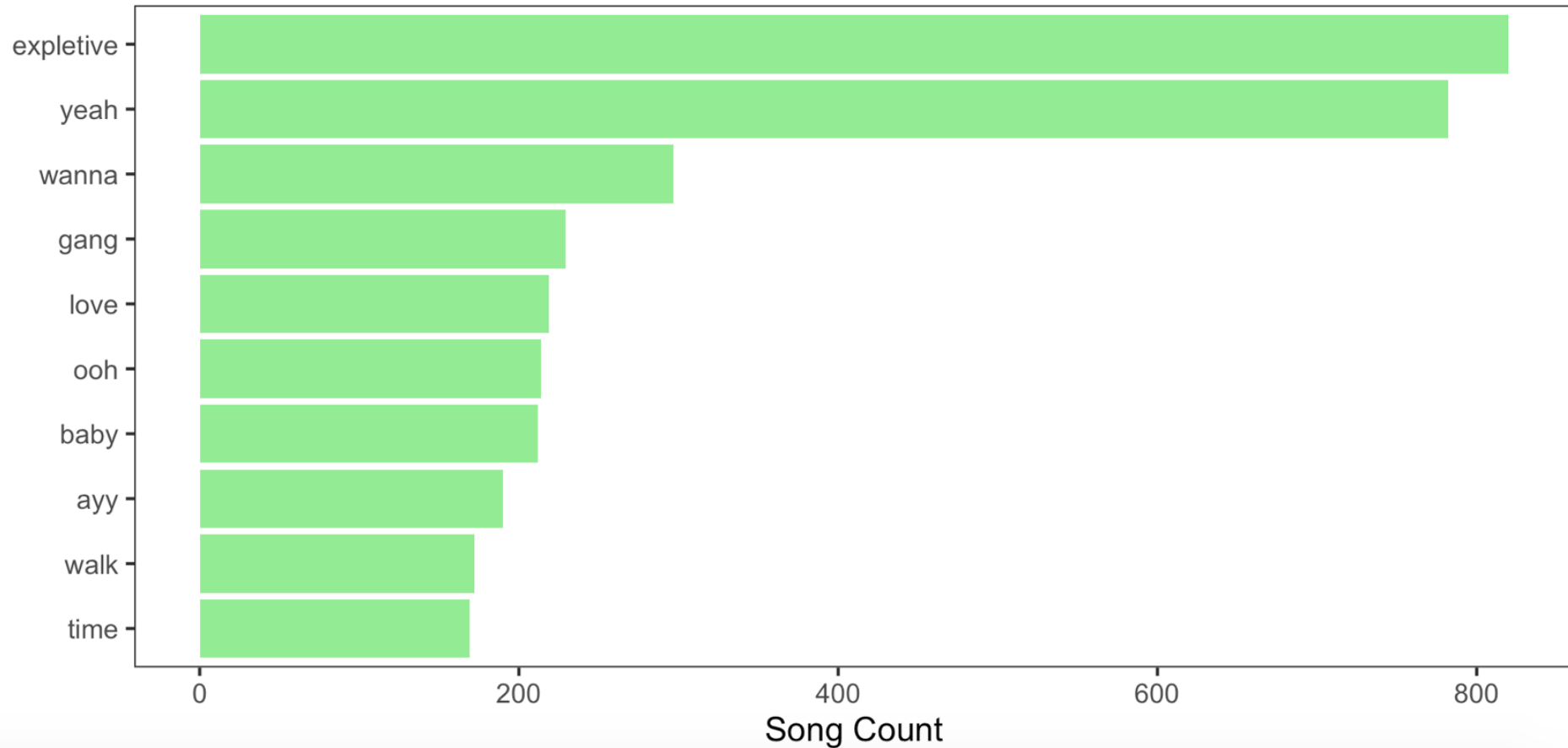
Goal: to use TF-IDF to *find the important words* for the content of each document by decreasing the weight for commonly used words and increasing the weight for words that are not used very much in a collection or corpus of documents

Calculating TF-IDF attempts to find the words that are important (i.e., common) in a text, but not *too* common

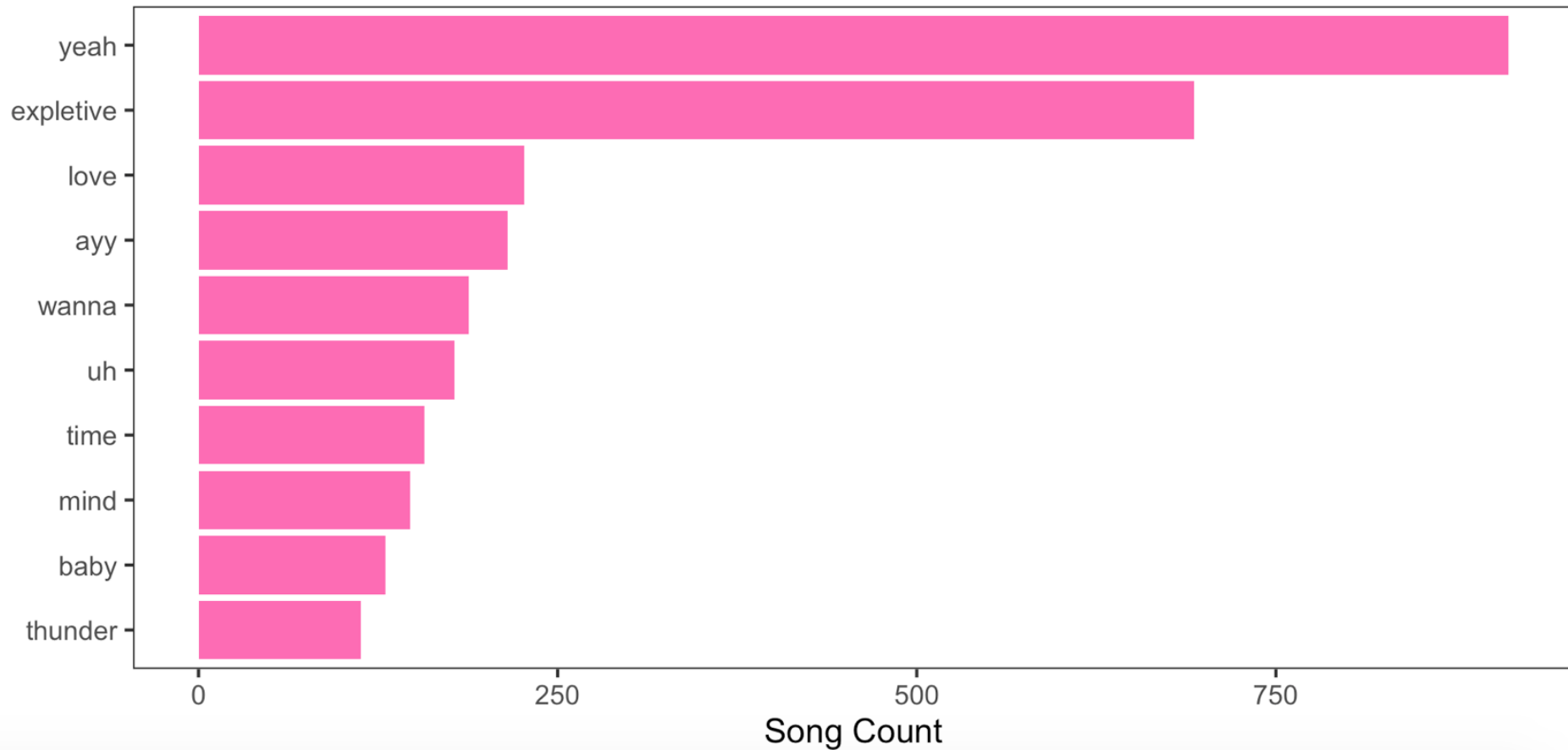
Most Frequently Used Words in top 200 songs (2017)



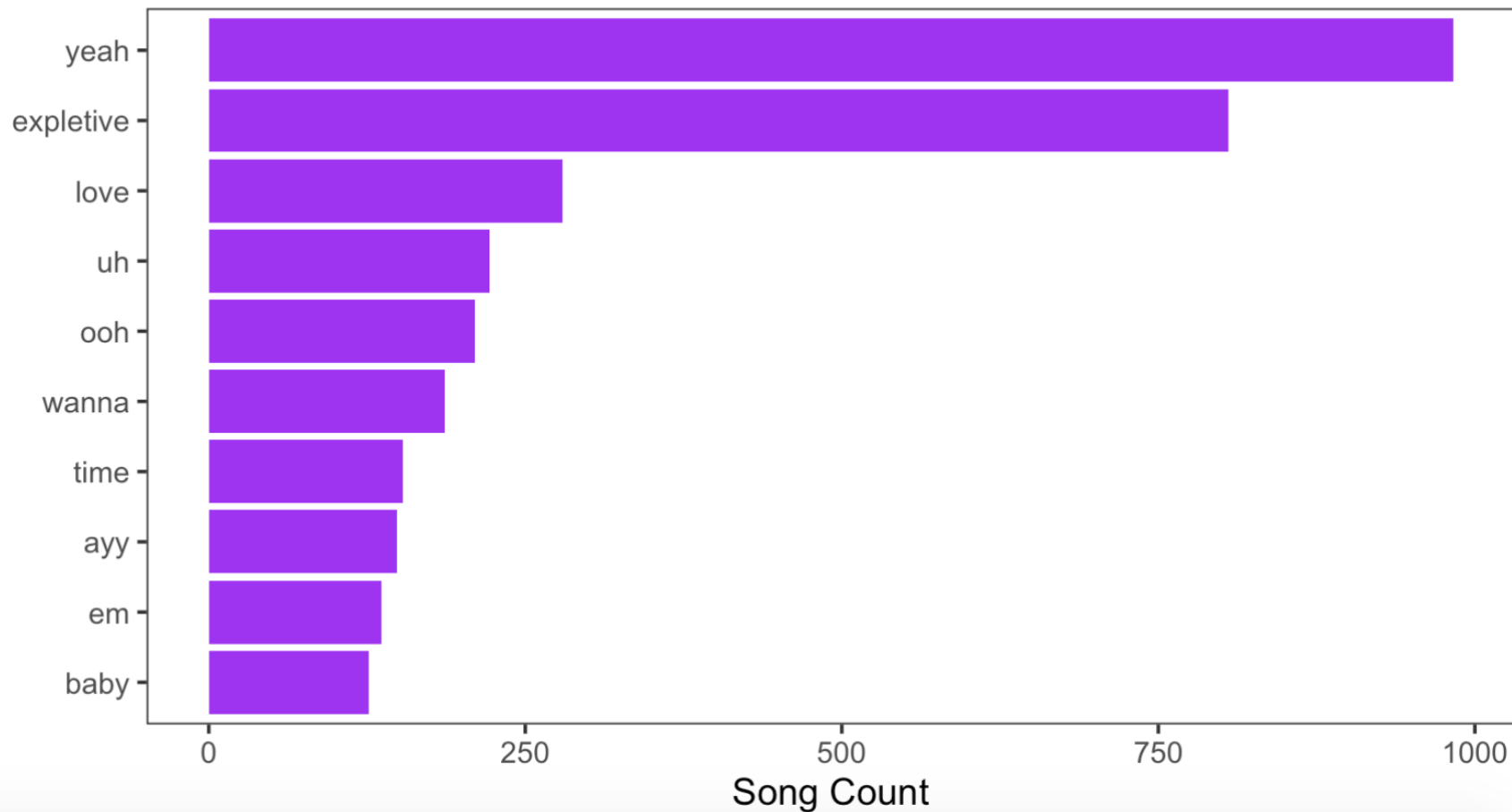
Most Frequently Used Words in top 200 songs (2018)



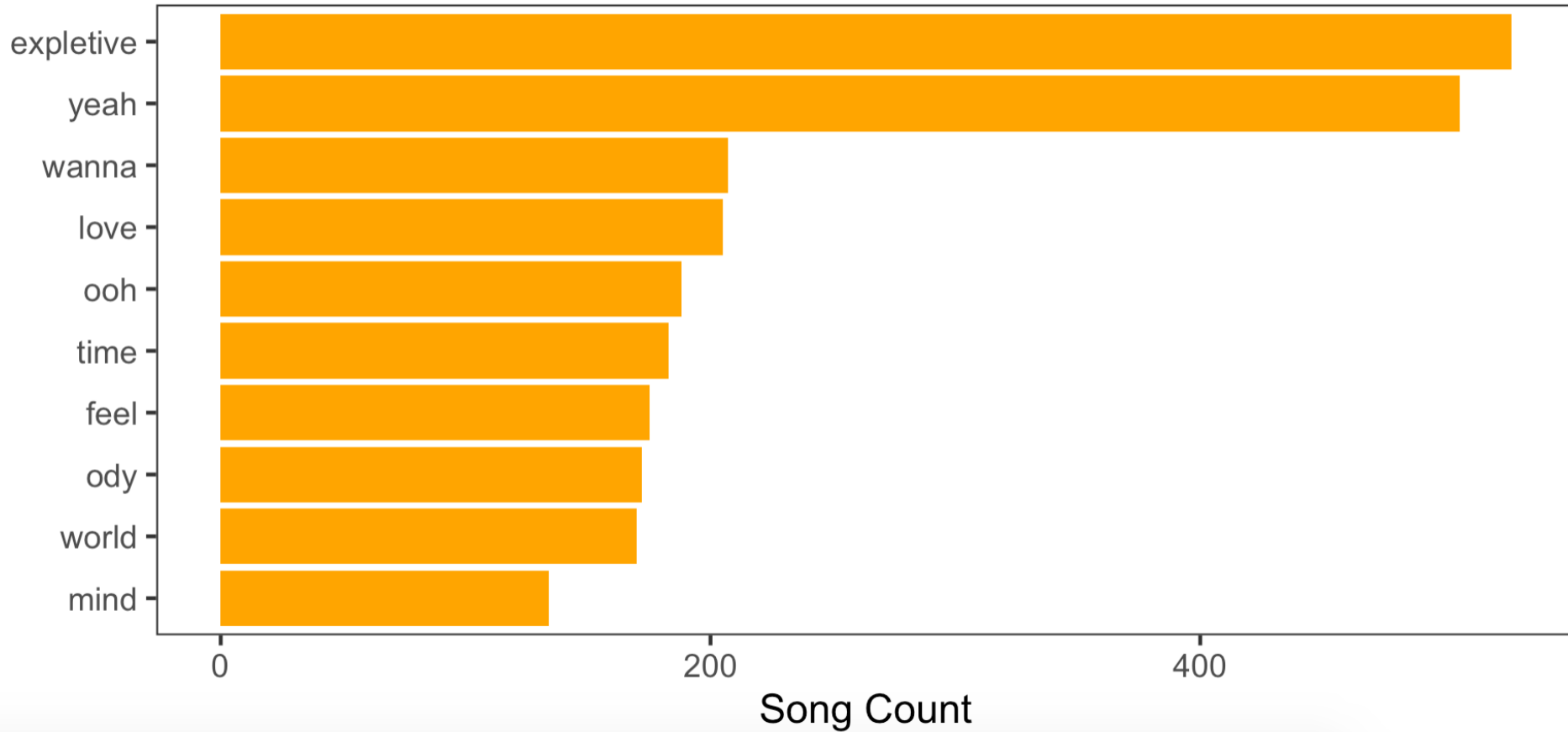
Most Frequently Used Words in top 200 songs (2019)

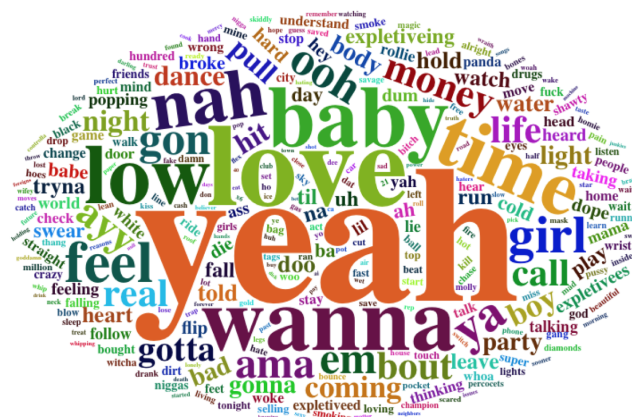


Most Frequently Used Words in top 200 songs (2020)



Most Frequently Used Words in top 200 songs (2021)

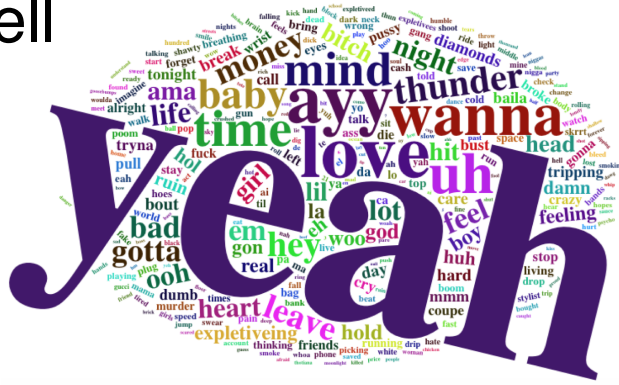




2017



2018



2019



2020

Term
Frequency
can only tell
us so
much....



2021

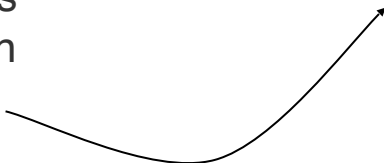
TF-IDF:

Term Frequency - Inverse Document Frequency

Term Frequency (TF) : how frequently a word occurs in a document

Inverse document frequency (IDF) : intended to measure how important a word is to a document

decreases the weight for
commonly used words and
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TF-IDF

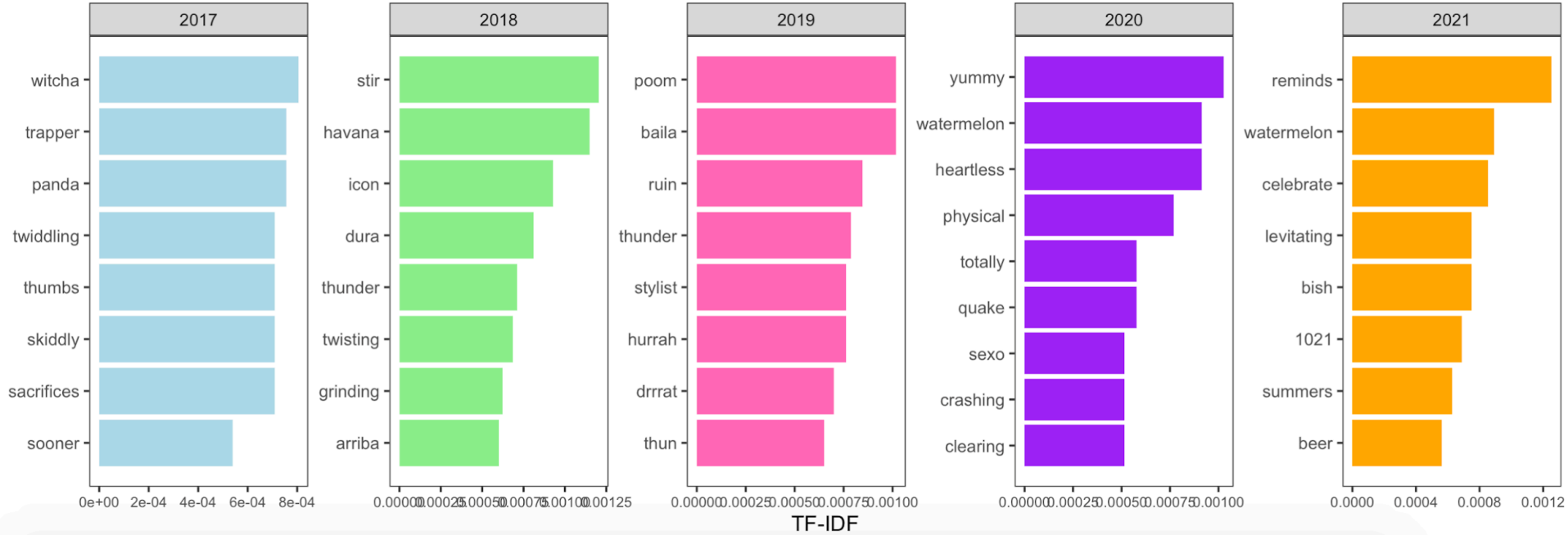
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Important Words using TF-IDF by Year



Father and Son

Flight of the Conchords



Live in London

2019

Just you and I
Driving round in the car
I even let you drive
Eating dinner from a can
dad's **yummy** can delight

I'm Me

Chai



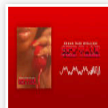
Punk

2019

ビーフ アンド チャオズ フィッシュ オレンジ
Everything **yummy** foods!

Sex Talk

Megan Thee Stallion

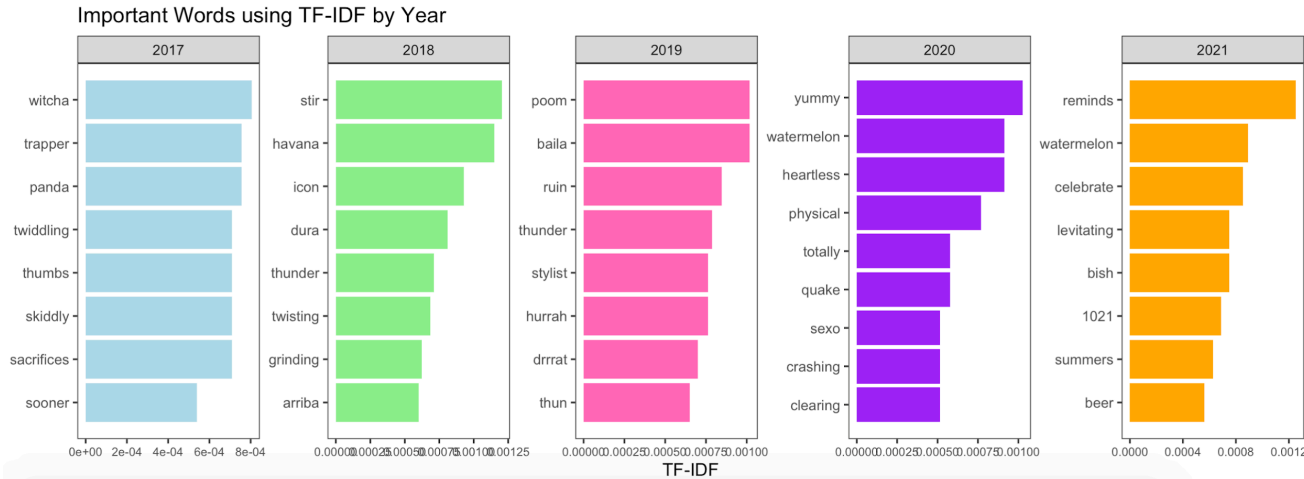


Sex Talk

2019

He eat from the back to the front
Bitch I got green like I'm Buttercup (ah)
He say I got that lil **yummy** yum (ah ah)

What can you conclude from this TF-IDF plot?



A No words overlap across the years in these data

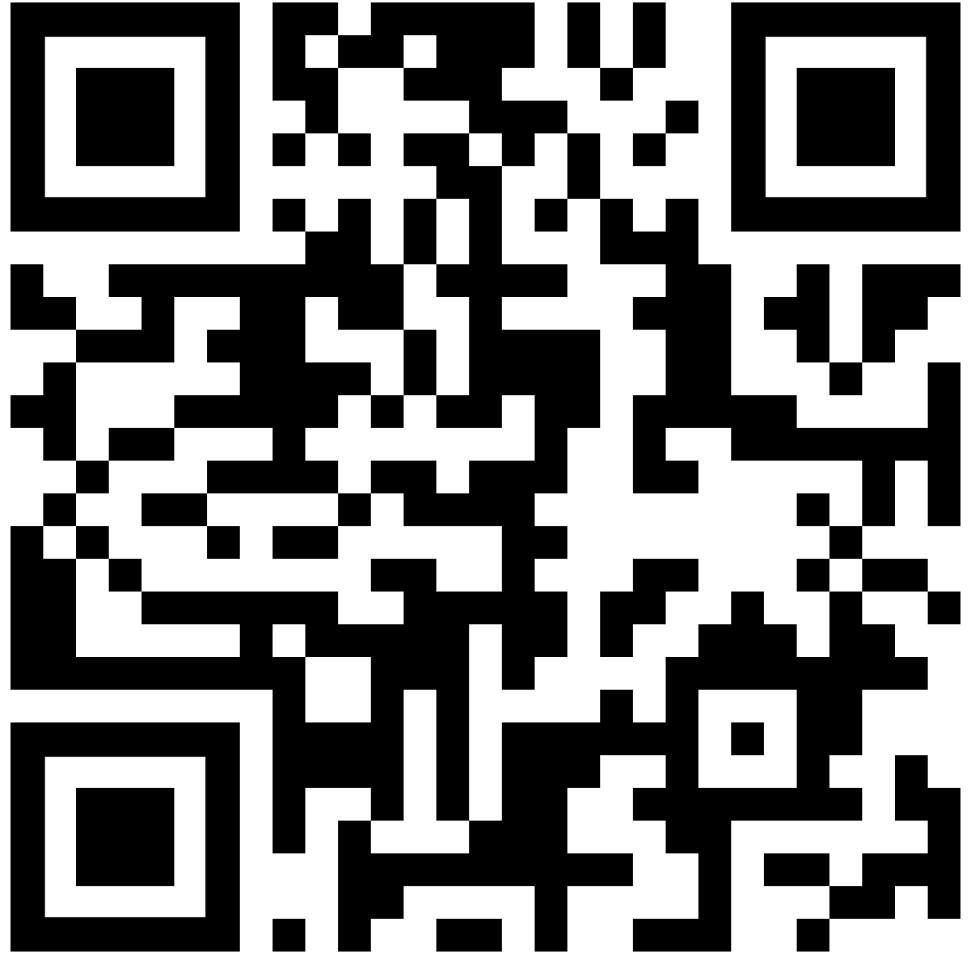
B 'reminds' and 'watermelon' are the most unique words to the 2021 data

C 'watermelon' is the most common word in this dataset

D A-C (all of the above)

E None of the above

<https://forms.gle/uD8cVTUUzAsxcVvKA>



Questions we can ask...

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EDA

TF-IDF

Sentiment
Analysis

Text analysis

The big picture

- Identify the problem you are trying to solve... what you want to answer suggests the approaches you will use below
- Clean and preprocess the data
- Linguistic transformations
 - Tokenization, lemmatization, stemming, remove stop words
 - [optional] part of speech tagging, named entity recognition
- EDA
- [optional] Calculate measurements and statistics
 - e.g., diversity, density, sentiment, TF-IDF
- [optional] Transform representation
 - Bag of words, vector embeddings
- [optional] Train a ML system or write an algorithm to accomplish a task given the representation
 - e.g., detect spam